1. Let
$$f(t) = t^3 + t + 1, \qquad g(t) = t^2 + t + 1.$$
Compute explicitly in $\mathbb{F}_2[t]$:
$$f + g,$$
$$fg,$$
and
$$f^2.$$
Write the final answers with coefficients reduced in $\mathbb{F}_2$.

2. Let F be a field. Determine all units of
$$F[t].$$
Prove your claim. You may use
$$\deg(fg) = \deg f + \deg g$$
for nonzero polynomials.

3. List all monic quadratic polynomials
$$f \in \mathbb{F}_2[t].$$
For each one, determine whether it is reducible or irreducible. You may use the root criterion for quadratics. Be explicit about which elements of $\mathbb{F}_2$ must be checked.

4. Let
$$f, g \in F[t]$$
be nonzero polynomials and suppose
$$f \mid g.$$
Prove that
$$\deg f \le \deg g.$$
Then determine what additional conclusion follows if
$$\deg f = \deg g.$$
Do not assume the second conclusion; derive it from the definition of divisibility and the degree formula.

5. Consider the two statements:
$$6 = 2 \cdot 3$$
in $\mathbb{Z}$, and
$$t^2 + t = t(t+1)$$
in $\mathbb{F}_2[t]$. Explain precisely, using the definitions of divisibility and irreducibility, in what sense
$$t \quad \text{and} \quad t+1$$
play roles analogous to prime factors. Then answer: Is $t^2 + t + 1$ irreducible over $\mathbb{F}_2$?
Give a proof, not merely a factor search.